PayLater Automation

Based on the user profile we need to grant him a dedicated limit of loan. Here is an idea to make this process automated.

1. Upload Required Documents
    a. Salary Certificate (pdf)
    b. Bank Statement (Salary Account only and pdf)

2. Document verification
    a. Original Document
        i. Pdf Digital Signature
        Signature metadata status we can check to keep either document is real or not. To do this we can use Python Libraries(**pyPDF2, pdfminer, pikepdf**). This can help us to check signatures and change history.

        ii. Pattern Layout
        Pdf layout which include header, footer and column alignment. With the help of **OCR** we can verify whether it is real or not.

        iii. Digital pdfs have vector text extraction so we ccan verify that.

    b. Forgery Detection
        i. AI Vision model (**openCv, PIL**) used to check it is change not modification in pdf or (image base pdf).
        ii. LLM model for text analysis consistency of words.

    c. Cross Match
        i. Names matching (Account, employee)
        ii. Compare monthly deposit salary matched with bank statement and salary certificate
        iii. Deposit amount frequency match to months (28 to 31 days)

NOTE: we can assign value to each step and keep the final weights of the document to mean how much is real. So based on that we can use this value for the grant limit.

3. Convert into Structure Data
    a. Monthly fixed salary
    b. Allowances
    c. Deductions
    d. Net Income
    e. Expenses (From bank statement flow of cash)
    f. Salary deposit date (from bank statement for keeping repayment)

4. Grant Limit
    a. Give information to Model (to go below section for more details)
    b. Process (Model result + Policy rules + Risk factor)
    c. Generate Results
        i. Give final limit to user and show in his profile(dashboard)

5. Repayment
    a. Keep record of statements
    b. Scheduled for repayments

Model To Process Limit

We need to train a model that will take user salary and statement records and give us a limit to grant.

1. Train model with data set
2. Feature engineering
    a. Net income (salary certificate)
    b. Debit ratio (bank statement)
    c. Credit ratio (bank statement)
    d. Saving ratio /stability (debit/credit ratio)
    e. Employee history /(bank statement)
    f. Document validation ratio (from above steps in document verification)
    g. Account activation time (bank statement)
    h. Limit (loan limit granted base on all above params)

3. Prediction (ML Model) & Generate results
    a. Loan limit with repayment schedule
    b. Risk factor

STACK

1. OCR + Paring Layer
    a. Pdfminer, PyPdf2 (for metadata & signature)
    b. PyMuPDDDF (Text Extraction)
2. ML Feature Engineering
    a. Pandas, scikit-learn
3. Fraud and Forgery Detection
    a. OpenCv, PIL